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Reducing Supply Losses in Vessel Oil Transportation in Commercial & Trading Sub-Holding, Indonesia Oil SOE Through an AI-Enhanced Digital Follow-Up Platform: INOVASI on Supply Loss (IOS)

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Abstract

"INOVASI On Supply Loss" (IOS) platform, an AI-enhanced digital follow-up solution management platform that we designed to significantly mitigate Over Tolerable Losses (OTL) in vessel oil and gas transportation at Pertamina, Indonesian State Oil Enterprise (SOE). Primarily targeting Pertamina's Commercial & Trading (C&T) Sub-holding, the IOS platform replaces traditional, delayed manual monitoring with an automated, near real-time detection and resolution system. Leveraging both pre-defined Subject Matter Expert (SME) rules and Artificial Intelligence (AI) analytics, IOS efficiently identifies supply transaction anomalies exceeding critical thresholds (e.g., loading loss (R1) and supply loss (R4) above 0.125%), initiating immediate alerts to responsible personnel.

The IOS platform includes an integrated workflow management feature that systematically manages the entire incident resolutions process, from reporting to corrective actions, offering comprehensive performance metrics and incident status monitoring through an AI-supported dashboard to significantly enhancing response times and operational transparency. The results from implementing IOS at C&T Sub-holding have demonstrated substantial improvements, notably a reduction in year-to-date supply loss from 0.125% to 0.07%, substantially exceeding both internal KPI (0.125%) and international benchmarks (0.3%-0.5%). The platform facilitated a verified cost avoidance of USD 35.32 million by December 2024. Furthermore, IOS eliminated false alarms entirely, ensuring 100% accuracy in anomaly detection, and notably increased incident resolution rates from approximately 39% to 85% within the C&T Sub-holding post-implementation.

The IOS platform represents strategic integration of AI-driven technologies specifically designated to detect, manage, and mitigate supply losses in custody transfer operations via marine vessels. Its innovative combination of Subject Matter Expert (SME) defined rules, AI analytics, and automated workflow management within low-code infrastructure delivers a scalable, data driven solution that surpasses traditional manual methodologies and setting new industry standards for efficiency and compliance.

The approach has proven effective in significantly reducing operational losses and achieving substantial financial savings. The initiative was successfully deployed in Commercial & Trading (C&T) Sub-holding and then replicated within Pertamina's Refining & Petrochemical (R&P) Sub-holding. This success demonstrates its scalability and significant potential to enhance financial, environmental, and governance outcomes across the broader oil and gas sector.

Introduction

Over the past several years, Indonesia's state-owned oil company has faced ongoing challenges in minimizing **supply losses** during vessel-based fuel transfer operations. In 2023, total losses of crude and refined products amounted to approximately **0.07% of all marine transactions**. While this figure remained within internal performance thresholds, it highlighted opportunities for efficiency enhancement. The Commercial & Trading (C&T) sub-holding alone accounted for roughly 65% of these losses. Such losses not only effect financial performance but also raise concerns regarding operational efficiency, environmental impact, and the potential for fraud. Contributing factors included the vast scale of operations – over *1,500 vessel calls per month* across more than *100 fuel terminals* nationwide – and **lagging manual processes** for monitoring and follow-up. Under the manual system, incidents of **Over Tolerable Losses (OTL)** were frequently identified only during periodic reviews conducted week or months later, allowing issues to persist unchecked. These circumstances clearly demonstrated the need for a more robust and effective control mechanism to enhance oversight and drive operational excellence in loss management.

To tackle this problem, the company developed an **AI-enhanced digital follow-up platform** using **Manage by exception and integrated workflow management concept** known as **INOVASI On Supply Loss (IOS)**. IOS is a low-code/no-code platform designed to automatically detect anomalous losses in near real-time and trigger immediate action. It integrates rule-based and AI-driven analytics to flag irregularities, sends instant notifications to responsible personnel, and manages the workflow of investigations through resolution – all in a centralized system. *"IOS is not just a dashboard, but a framework for strategic, data-driven supply loss control with a Plan-Do-Check-Act cycle,"* as one project lead described. By moving the process from spreadsheets, emails and after-the-fact reports to an automated, unified platform, IOS represented a **breakthrough change from manual to digital** operations. Early results have been very promising: supply loss incidents are now identified and responded to much faster, the overall loss rate has started trending downward, and the company has realized substantial cost avoidance – on the order of **USD 35.32 million** saved in a year. This manuscript details the challenges in vessel oil transfer operations, the design and implementation of the IOS platform, and the outcomes achieved. It also discusses how IOS aligns with broader organizational goals (including safety, sustainability, and governance) and the lessons learned in driving digital transformation in a large oil & gas enterprise.

Background and Problem Statement

Supply loss refers to the discrepancy between the volume of oil products loaded versus unloaded during transfer operations – essentially, the lost product that cannot be accounted for due to factors like evaporation, measurement inaccuracies, or pilferage. Every company sets an acceptable tolerance for such losses (often a percentage of total throughput), and any loss exceeding that threshold is considered an OTL (Over Tolerable Loss) event. In the Indonesian oil SOE's case, the acceptable loss percentage is quite low (on the order of 0.1%), reflecting high standards. Despite this, the absolute value of losses had been rising since 2020, peaking in 2023 as noted. The C&T sub-holding responsible for nationwide fuel distribution via marine transportation, was the primary contributor to these losses. Its operations are extensive, involving hundreds of ships voyages each month to deliver fuel to depots across the Indonesian archipelago – constituting one of the most complex supply chains in the world. This operational complexity posed significant challenges to

effective loss monitoring. Subsequent investigations identified several critical pain points that contributed to the uncontrolled losses:

- **High Volume of Shipments:** With 1,300–1,500 ship calls per month in the C&T segment, the transaction volume is enormous. Manually tracking each transfer's gains or losses in near real time was practically impossible under the old system.
- **Geographically Dispersed Terminals:** The company operates **100+ fuel terminals** across Indonesia. Communication and coordination across distant sites posed a challenge, often resulting in delayed reporting of anomalies.
- **Outdated Monitoring Process:** The existing monitoring mechanism for OTL incidents – known internally as DNA-OTL (Daily Notification Awareness – Over Tolerable Loss) – was largely **manual and lagging**. Field staff would compile loss events and email reports, and a centralized team would review them in quarterly workshops. This lag meant that by the time an issue was flagged (e.g. a particular port consistently losing product beyond tolerance), significant time had passed. Opportunities to intervene early or prevent recurrence were missed.
- **False Alarms and Data Issues:** The manual system often generated **false alarms**, and its criteria were basic. Small measurement errors could trigger alerts that did not truly indicate a problem, causing "alarm fatigue" and reducing trust in the system. Conversely, some genuine issues might have been overlooked due to crude rules.
- **Slow Response and Closure:** Even when an OTL incident was identified, the follow-up process was cumbersome. Preparing the official **Loss-Event Report (LER)** – essentially an incident report – required gathering data and approvals through paperwork, which **took a long time**. Meanwhile, actions to address the loss (e.g. investigating for leaks or pilferage, recalibrating meters, etc.) were often delayed or not thoroughly documented. Prior to IOS, only about **39% of reported OTL incidents in C&T** were ever formally closed with an LER and resolution, indicating many cases remained open or unresolved.

These issues underscored the need for a **tighter control mechanism** and a shift toward proactive management of supply losses. The situation not only had financial implications but also **Sustainability (ESG) impacts**: environmentally, higher losses meant more hydrocarbon emissions (vapor loss or spills) and wasted resources; socially, unchecked losses could hint at fraud or negligence, eroding stakeholder trust; and in terms of governance, it signaled non-compliance with internal controls and policies. In short, improving supply loss performance was critical for operational excellence, cost optimization, and upholding the company's reputation and governance standards. By late 2023, the leadership set a mandate to drastically reduce supply losses in C&T and enterprise-wide, as part of a broader drive toward efficiency and accountability.

Solution: The IOS Digital Platform for Loss Control

To address the above challenges, the team implemented a comprehensive digital solution: **INOVASI On Supply Loss (IOS)**. The IOS platform was built on **INOVASI**, an in-house low-code/no-code application framework, allowing rapid development with minimal programming. At its core, IOS serves as an **AI-enhanced anomaly detection and workflow system** dedicated to supply loss events. The solution concept was to automate the entire Plan-Do-Check-Act cycle of loss control:

- **Detection (Plan & Do):** IOS continuously monitors fuel transfer data (from flow meters, tank gauges, bills of lading, etc.) and applies predefined **rules and machine learning** to detect anomalies. *"IOS can detect anomalies or outliers based on preset rules,"* explained the SVP of the Pertamina Digital Hub. Domain experts defined what constitutes tolerable loss for each route and product; any deviations beyond those thresholds trigger what is called an **exception signal**.

Machine learning algorithms were also introduced to identify patterns (for instance, comparing losses across similar voyages or detecting when multiple small losses might indicate a bigger trend). This reduces false alarms by using context – for example, accounting for weather or voyage duration – and catches subtle issues that simple thresholds might miss.

- **Alert & Notification (Do):** Once an OTL exception signal is generated, IOS automatically issues an **alert**. The relevant person-in-charge (PIC) at the affected terminal or operation receives a notification (through email). This happens in near real-time, often immediately after the unloading reconciliation is done. Previously, operators might not learn of an issue until an end-of-month report; now they are alerted instantly. One field operator noted that with IOS, *"we get notified of anomalies as soon as they occur, which allows us to act before the next transfer – previously, we might not hear about it until weeks later."* (This is based on multiple operator testimonials compiled during the IOS pilot, which consistently highlighted improved response speed)
- **Integrated Workflow (Act):** Upon notification, IOS invokes a **GenAI-assisted categorization service** that enriches each exception signal with structured tags (event type, suspected cause, asset/equipment class, operating phase, severity, and data-quality confidence) and generates a draft Loss Event Report (LER). The service also clusters related signals across voyages/assets within the same processing window. It operates in periodic micro-batches (e.g., end-of-day/week) to triage cases at scale while avoiding duplicate investigations. Categorized cases are automatically routed to the appropriate Person-in-Charge (PIC) via policy rules (by product, route, or asset class) supported by **retrieval-augmented playbook suggestions** for corrective action. Escalation and Service Level Agreement (SLA) logic remain intact; time-bound steps, reminders, and management escalations are applied per category and risk level, ensuring a single auditable trail. This approach reduces operator overload by prioritizing only the highest-impact clusters. This **workflow management** aspect ensures that every loss incident is *"tracked to closure in near real-time and with full transparency"*. It effectively replaces ad-hoc emails and Excel trackers with a structured, automated process.
- **Reporting & Monitoring (Check):** IOS provides a **central dashboard** that aggregates all OTL signals and their status. Managers at various levels can see the number of open cases, those resolved (closed with LER), and key performance metrics such as average closure time and loss reduction achieved. The dashboard updates continuously as data comes in, and actions are taken, providing a **near real-time view of supply loss management**. Each closed case carries a validated **quantity-accounting impact** (book-to-physical variance, percentage loss and absolute volume) which is rolled up to the corporate level. This allows management to track **value creation measured based on quantity accounting** (e.g., avoided loss, recoveries) alongside operational KPIs (Integrated Loss Control/ILC). Compared with the previous quarterly static reports, this represents a significant improvement in timeliness and transparency. In addition, periodic summary reports (e.g., a monthly summary of all losses and actions) are generated to inform senior management and feed corporate risk reviews. The integrated dashboard and automated reporting substantially reduce manual administrative work; one could say the **"full digitalization of what used to be a tedious manual recap process"** has freed up staff to focus on analysis and decision-making rather than paperwork.

The Continuous Improvement of Supply Loss Control

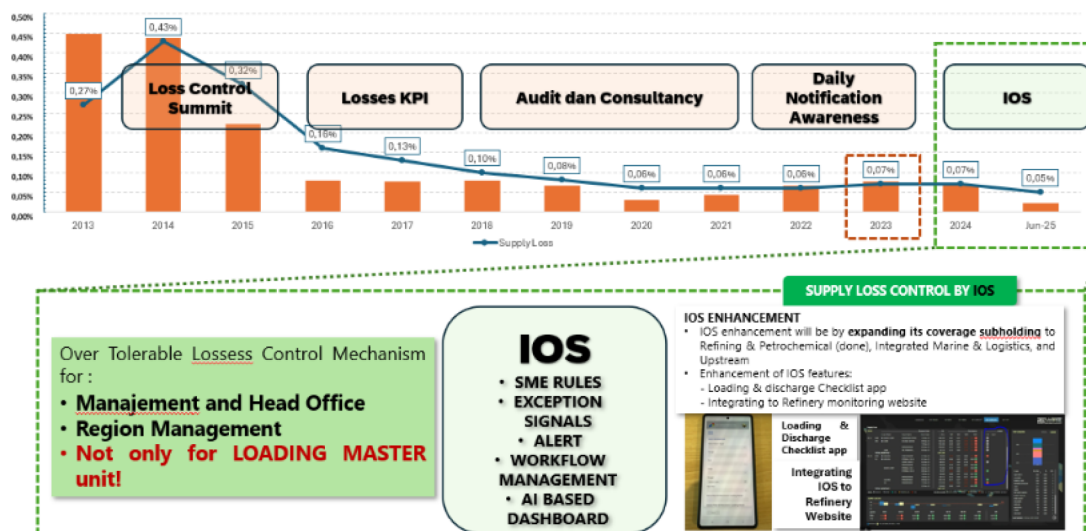


Figure 1—Historical trend of corporate supply loss (2013–2025) and the impact of IOS implementation. The blue line (left axis) shows supply loss as a percentage of total throughput, which peaked in 2023 and dropped in 2025 after IOS rollout (from ~0.07% to ~0.06% and 0.05 % June 2025 (YTD)). The introduction of IOS in 2024 created a cost avoidance of USD 35.32 million (relative to the business plan target), highlighted in green. This value creation, achieved at minimal cost, was formally validated by the company's Internal Audit department as a major cost optimization success.

Platform Design and Capabilities

Technically, IOS leverages the **INOVASI** platform (short for *INtegrated OPTimization based on VALuable SIGNAL*). INOVASI provided a ready-made infrastructure with modules for data integration, rule engine, notification service, and workflow designer, which accelerated development. The team configured INOVASI to suit supply loss management needs. Key capabilities of the IOS platform include:

- Rule-Based & AI Detection:** A hybrid approach was used for anomaly detection. *Business rules* (defined by subject matter experts) capture known conditions for losses – for example, triggering an *OTL signal* if *loss > 0.3% for a short-haul voyage*, or if *the absolute loss > X kiloliters*. These rules ensure that critical thresholds are never overlooked. In parallel, machine learning algorithm learn expected behaviors and detect outliers and latent patterns (e.g., a route-vessel pair trending abnormally even when single events remain below threshold). Generative AI (Large Language Model, LLM) components further normalize heterogeneous signal payloads and auto-classify probable causes, thereby improving both sensitivity and specificity. For instance, the AI can flag when a particular vessel on a certain route consistently shows higher than expected losses, even if each single event is just under the defined threshold. By combining these approaches, IOS improves accuracy and reduces false alarms.
- Automatic Notifications:** IOS is integrated with the company's email system, so that when an OTL event is detected, it **automatically sends notifications** to designated personnel. Each OTL signal is tagged a responsible party (e.g. a regional supervisor or a terminal manager). Notifications include basic details of the event and a link to the IOS application for further action. This ensures **no time is lost** in alerting – a vast improvement over the previous manual email chain, where an alert might sit unnoticed in an inbox. According to internal records, the average time from loss *to the first alert has fallen from days (or the next scheduled report) to almost instantaneous now* (within minutes) after IOS.
- Integrated Workflow Management:** The platform's built-in workflow engine was configured to model the **end-to-end follow-up process** for any loss incident. It starts with the initial

acknowledgment of the incident by the Person-in-Charge (PIC), followed by root cause analysis, action assignment (such as initiating equipment checks, recalibrations, or fraud investigation), and concludes with the approval and closure of the LER. This workflow is **fully digital** – approvals are performed online, and all stakeholders have near real-time visibility of status updates. The IOS workflow enforces accountability. The PIC validates volumes and causes; finance reviewers confirm posting periods and general ledger mapping; upon approval, entries are automatically pushed to corporate consolidation. Escalations are prioritized based on both operational risk and value-at-stake, ensuring attention is focused where enterprise impact is greatest. This standardized digital workflow enables scalability across terminals, fleets, and regions, while making value creation—measured via quantity accounting—visible and comparable company-wide. This has led to a more disciplined approach in handling losses. According to a senior operations manager, tasks that previously *"took weeks of back-and-forth are now completed in days because the system keeps everyone on their toes."*

- Dashboard and Analytics:** A unified dashboard provides near real-time visibility of incidents and performance with filters aligned to corporate indicators (throughput, loss %, cost per kiloliter). The dashboard offers multiple views, including variance waterfalls (from detected signals to approved postings), trend analyses by asset/route/product, and value-creation metrics (e.g., loss avoided, recovery realized, cycle time to posting) derived from consolidated quantity-accounting records. Cross-asset normalization and drill-downs capabilities deliver decision-ready insights while reinforcing transparency and accountability at scale (as illustrated in Figure 2). Data from IOS is now actively used in management forums – for example, monthly OTL workshops between the holding and sub-holdings leverage IOS data directly, replacing static slide presentations. Furthermore, the data has been integrated with other systems, such as the Refinery Monitoring Excellence (REMOTE) platform, to enable broader cross-functional analysis.



Figure 2—INOVASI on Supply Loss (IOS) Dashboard Summary Sub-Holding C&T.

Notably, IOS was developed and implemented **entirely in-house** by a cross-functional team from the Integrated Loss Control group, Pertamina Digital Hub and the C&T sub-holding itself. The project leveraged

internal expertise and existing infrastructure, resulting in a remarkably low implementation cost of only **IDR 50 million** (approximately **USD 3,300**). This cost mainly covered some workshops and user training sessions; no expensive new software needed to be purchased, since the INOVASI platform was already available. The in-house approach also ensured that the solution was tailored to the company's processes and that knowledge was kept internally for future improvements.

Implementation and Change Management

Rolling out IOS required careful attention to change management, given it touched many parts of the organization. The implementation proceeded in stages:

1. **Pilot Phase:** The team first piloted IOS in one region (North Sumatra area) in Feb 2024. This allowed testing the system on a smaller scale and collecting user feedback. The pilot involved training the local terminal staff and loss control PICs on how to use the platform and interpret the signals. Initial pilot results were positive – within a couple of months, the region saw a reduction in repetitive loss incidents and much faster closing of cases. Encouraged by this, management gave a green light for wider deployment.
2. **Enterprise Deployment:** In mid-2024, IOS was rolled out across all regions of the C&T sub-holding. A series of **training workshops** and socialization sessions were conducted for terminal managers, shipping coordinators, and relevant headquarter analysts. Because IOS was designed with a user-friendly web interface and integration into familiar tools such as email, adoption was smooth and efficient. Additional support was provided via a helpdesk and a user guide. The project team emphasized that IOS was not intended to "police" the field operation, but rather to **empower** personnel with better tools – an important message in securing user buy-in. This framing proved effective; many field staff quickly recognized IOS as a helpful assistant rather than a burden, particularly as it automated repetitive and time-consuming tasks.
3. **Monitoring & Support:** After go-live, the team established a rigorous monitoring and support routine. They held **weekly/biweekly check-in meetings** for the first two months after deployment, gathering all regional representatives via online to discuss any issues, gather feedback, and ensure proper use of IOS. Additionally, **daily monitoring** was set up via the dashboard – headquarters could see every day if new signals were coming in and being handled. A centralized log tracked any application glitches or user difficulties, which were addressed promptly by the support team. This close support during the early phase was crucial to driving adoption and demonstrating quick wins.
4. **Formal Evaluation & Integration:** By 3rd qtr-2024, an evaluation of IOS's impact was carried out. The results were compiled and presented to senior management in a formal report (including cost avoidance figures, reduction in OTL events, etc.). The evaluation also included a **user satisfaction survey** administered by a third party to ensure objectivity. Impressively, over **83% of users** responded that they were satisfied or very satisfied with IOS, exceeding the 80% satisfaction target. Many qualitative comments praised the faster response times and ease of use. Some suggestions (like additional filters on the dashboard or clarifying certain alert messages) were collected for the next update. The positive results of this assessment strengthened management support and paved the way for expansion to other sub-holdings.

Throughout implementation, **change management** best practices were applied. The team secured a **formal mandate (policy)** for using IOS: a new technical guideline was issued, making the use of IOS mandatory for monitoring OTL incidents in C&T. This ensured everyone treated IOS as the official system, not an optional tool. Performance indicators were also adjusted – for instance, regional managers were now partially evaluated on their timely closure of IOS signals, injecting accountability. By embedding IOS into the standard operating procedures and KPIs, the changes were stickier.

Another key aspect was highlighting and **celebrating early successes**. When the first instances of loss avoidance due to IOS occurred (for example, a case where an alert led to discovery of a leaking flange and saved significant product), those stories were circulated in internal newsletters and town-halls. This built positive momentum and showed staff that the new system had yielded tangible results.

Results and Discussion

The implementation of IOS in the C&T sub-holding has led to marked improvements in supply loss control. [Table 1](#) summarizes some key metrics before vs. after IOS deployment, demonstrating the platform's impact on operational performance:

Table 1—Key Operational Metrics Before and After IOS Implementation (C&T Sub-holding)

Metric	Before IOS	After IOS
Total Supply Loss (2023)	Increasing trend	<i>Reduced in 2024 (trend ↓)</i>
C&T Share of Loss	65% of total supply loss	<i>Major reduction achieved (share↓)</i>
Supply Loss Rate (% of throughput)	Corporate ~0.07% (2023) C&T ~0.08% (2023)	Corporate~0.07% (2024); ~0.06% (Q1 2025) C&T ~0.07% (2024); ~0.06% (Q1 2025)
OTL Incident Response Time	Weeks to months (manual)	Near real-time (instant alerts)
LER Closure Rate (cases closed)	~39% (Jul 2024)	~86% (Q1 2025)
Cost Avoidance Realized	N/A (baseline only)	USD 35.32 million cost avoidance (2024)
Monitoring Frequency	Quarterly (lagging review)	Daily/Continuous (dashboard)
User Satisfaction	Low (manual burdens, delays)	> 83% satisfied (survey)

Before IOS figures refer to 2023 baseline or the last period without IOS; After IOS figures refer to late 2024/Q2 2025 performance with IOS in place. Arrows (↓) indicate downward trends (improvement) where exact figures are confidential but positive.

As shown above, the **percentage of product loss** improved from approximately 0.08% to 0.07% of total throughput. While the absolute change appears minimal, at the scale of hundreds of millions of barrels, it represents millions of dollars saved. Indeed, by meeting or exceeding loss reduction targets, the company avoided **USD 35.32 million** in cost avoidance. This figure was calculated by comparing the actual losses after IOS to the business plan's allowable losses and **validated by Internal Audit**. In fact, IOS paid for itself many times, given its negligible implementation cost.

One of the most dramatic improvements was in the **rate of closing OTL incidents**. Prior to IOS, only around 1/3 of identified loss cases in C&T were ever formally closed with an LER (the rest lingered or were quietly dropped). Now, with workflow enforcement, the closure rate in C&T surged to **85–86%** of cases. This means the vast majority of loss incidents are investigated and resolved. The remaining ~15% are usually those still in progress within the allowed time or awaiting parts for a fix, etc., rather than being forgotten. In other words, IOS instilled a culture of "**no loss event left behind**", which is a cornerstone of continuous improvement.

Another major benefit has been **response speed and accountability**. Field teams respond to IOS alerts in near real time – often mitigating an issue before the next transfer. For example, if a depot sees an unusual loss in a morning shipment, the team can inspect and recalibrate by afternoon, preventing repeat losses in the evening shipment. In the past, that depot might not realize the problem until the month's end report, by which time numerous faulty transfers could occur. This improvement in responsiveness is somewhat hard to quantify but was repeatedly highlighted in qualitative feedback. A manager from the Sumatra region described, "*What used to take weeks now is resolved in a day or two. We're catching problems as they happen, not after the fact.*" This proactive stance certainly contributed to the measured decline in losses.

From a **quality and compliance** standpoint, IOS ensures stricter adherence to control standards. The platform enforces the internal policy that every OTL must have an LER action plan, thereby strengthening governance. It also enabled better **fraud detection**; since data is centralized and transparent, any suspicious patterns (like consistent small losses at a particular terminal) are more easily spotted by headquarters. There have been cases where IOS data helped trigger audits or investigations into irregular activities, much less likely under the old system of dispersed records.

The intangible benefits are also noteworthy. The **efficiency gains** in administrative work have been significant – staff who previously spent hours every week collating loss figures and drafting reports have that work automated by IOS. One can imagine a morale boost when tedious tasks are eliminated. As one supervisor put it, *"IOS took away the drudgery of manual reporting, allowing us to focus on solving the problems instead of just reporting them."* This sentiment was echoed in the user satisfaction survey, which saw high scores particularly in areas of "usefulness" and "time saved."

The project also had positive **ESG implications**. Environmentally, reducing supply losses directly reduces hydrocarbon waste and emissions (less fuel lost = less vapor and spill risk). Socially, stronger control and transparency help prevent fraud or theft, improving trust in the supply chain. In terms of governance, IOS exemplifies compliance with and enhancement of internal controls (e.g., compliance to the internal loss control procedure). This is a good example of how digital innovation can support sustainability and governance goals in addition to financial goals.

Perhaps one of the most impressive validations of IOS's value is how quickly it attracted interest from other parts of the company. Seeing the success in the C&T sub-holding, the company's **Refining & Petrochemical (R&P) sub-holding** decided to adopt IOS. They went live in January 2025, and within one quarter, their results mirrored C&T's success: The R&P team's OTL closing rate leapt from a mere ~15% (pre-IOS) to **98%** after IOS implementation. Essentially all loss signals in R&P were being acted on and closed. This nearly perfect closure rate underscores that the platform and process can be effective across different operational contexts. Plans are now in motion to roll out IOS to the **Integrated Marine Logistics (IML)** sub-holding next (which handles vessel operations for crude and product transport) and even to the **Upstream** sub-holding for controlling crude loading losses. Such replication could multiply the benefits companywide. In addition, the team is already working on new features for IOS – for example, a digital **Loading & Discharge Checklist** to replace manual checklists during ship loading, which will further reduce human error and improve data capture. The IOS platform's data is also being fed into cross-entity workshops and other analytics tools (like REMOTE) to enhance enterprise-wide loss monitoring. These developments show that IOS has become a fundamental tool for **continuous improvement** in the company's loss control efforts.

Importantly, IOS changed the mindset around loss management. Before, losses were somewhat seen as an unfortunate but accepted cost of doing business, as long as they were under a threshold. Now, with near real-time data, every loss event is *visible* and actionable, fostering a culture of **accountability and problem-solving**. The initiative has thus contributed to building a more data-driven, responsive organizational culture. Management involvement has also increased—with dashboard and regular reports, senior leaders can ask questions about specific incidents, increasing attention on the issue.

To illustrate the qualitative impact, consider the testimony of one operations manager after using IOS for a few months: *"Implementing the IOS platform has been a strategic breakthrough that not only increases efficiency but also strengthens our control over supply losses. With automatic notifications and integrated exception signals, our response to anomalies is now much faster and more accurate, and processes that used to take a long time now run far more effectively, helping us keep losses below the set thresholds."* This sentiment captures how IOS transformed the loss control process from a reactive, tedious chore into a proactive, effective system.

Conclusion

The INOVASI On Supply Loss (IOS) platform has demonstrably transformed supply loss management in vessel-based oil transportation for the Commercial & Trading (C&T) Sub-holding, **reducing losses by 14.28% between 2023 and 2024** and delivering **validated cost avoidance of USD 35.32 million** in its first year. By replacing a lagging, manual monitoring system with a near real-time, AI-enhanced control framework built on a low-code platform with integrated workflows, the company shifted from reactive reporting to proactive, data-driven intervention. The results are clear: loss rates have declined significantly, incident resolution rates have surged from **~39% to over 85%**, and compliance, accountability, and response times have all improved.

The successful deployment in the C&T Sub-holding—followed by replication in the Refining & Petrochemical Sub-holding, where **closure rates increased from ~15% to 98%**—demonstrates both scalability and enterprise-wide impact. Beyond the measurable financial and operational benefits, IOS has provided a proven template for operational excellence, fully aligned with the company's strategic goals of efficiency, transparency, and continuous improvement.

The success of IOS offers a powerful template for operational excellence, yielding several critical lessons. The true innovation was not just its technology but its role as a catalyst for **process re-engineering and cultural change**. By embedding accountability into daily workflows through automated alerts and transparent dashboards, it fostered a culture of ownership and continuous improvement, moving beyond passive compliance. Developed internally on a low-code platform for a mere USD 3,300, IOS delivered a multi-million dollar return on investment, demonstrating that targeted, in-house digital solutions can be more effective and cost-efficient. Ultimately, the platform's greatest contribution was making every loss event visible and actionable in near real-time. This transparency empowered field operators to act immediately, armed management with data for strategic decisions, and strengthened governance by ensuring *"no loss event is left behind"*.

The journey of IOS is far from over; it serves as a cornerstone for future digital initiatives. The strategic roadmap is focused on evolving the platform from a specialized tool into a comprehensive operational risk management ecosystem. Immediate plans involve extending IOS to the Upstream and Integrated Marine Logistics (IML) sub-holdings to create a seamless, end-to-end view of hydrocarbon loss across the entire value chain. Future enhancements will focus on evolving the platform's intelligence from anomaly detection to prediction, enabling pre-emptive actions that prevent losses before they occur. The core INOVASI framework can also be adapted to address other critical operational challenges, such as demurrage or product quality incidents. In conclusion, the IOS platform has not only solved a critical business problem but has also forged **a new path for digital innovation**, setting new industry benchmarks for supply chain efficiency and risk management.